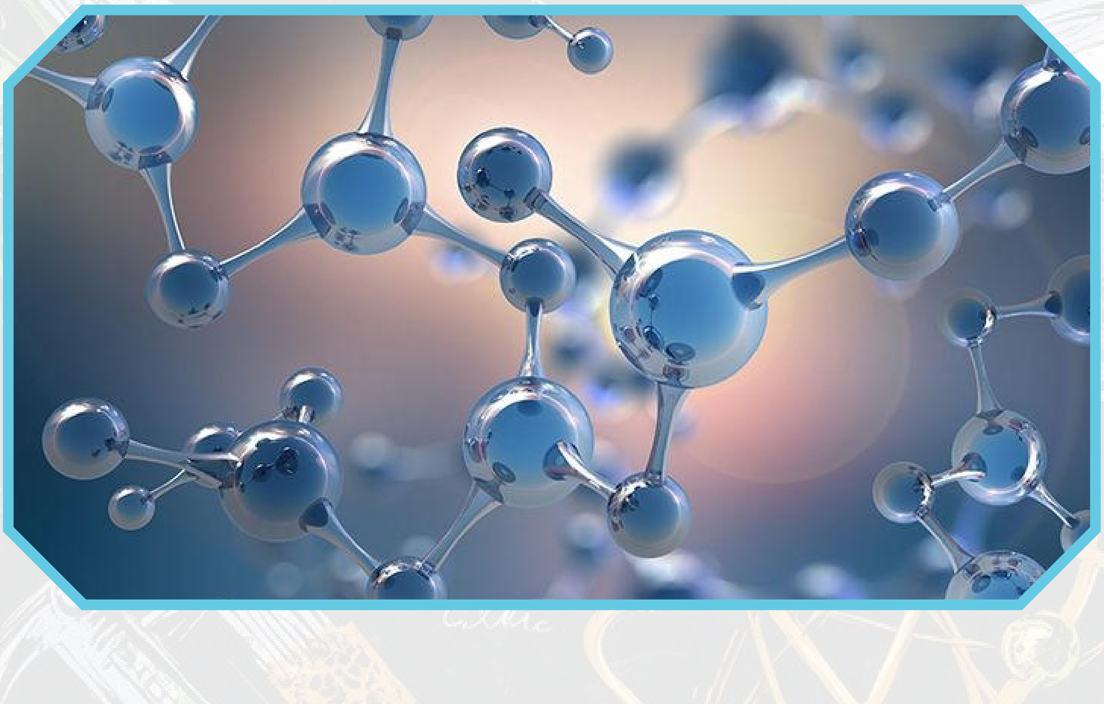
UNIT 23

DIGITAL ELECTRONICS

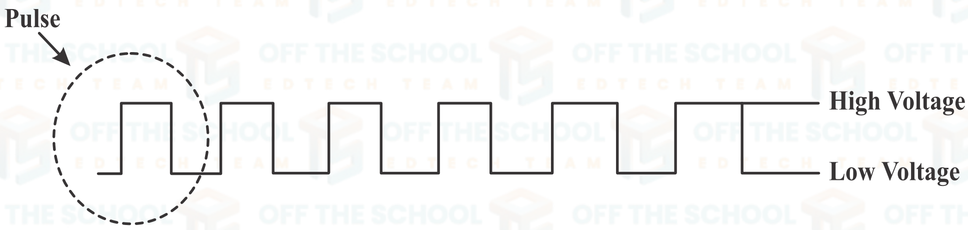


Introduction:
Digital electronics is a branch of electronics that deals with the manipulation of digital signals, represented as binary digits (0s and 1s).

Digital signal levels:
In digital electronics, signal levels are represented in various ways. Here are common digital signal levels:

1. Low Level ('0'): KNOW

• This represents the binary digit 0. Noise immunity is a.
• In terms of voltage, it is associated with a lower voltage measure of how well a level.
• Often referred to as "low" or "logic 0."



2. High Level (1): electrical signals or noise

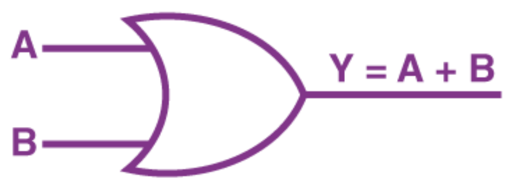
• This represents the binary digit 1, without reacting to them.
• In terms of voltage, it is associated with a higher voltage level.
• Often referred to as "high" or "logic 1."

Logic gates:
Logic gates are tiny electronic switches that enable decision-making in computers and other devices. They combine to perform logical operations, playing a crucial role in information processing.

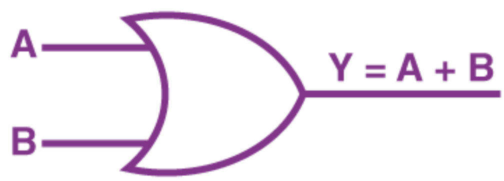
Types of Logic Gates:
Logic gates are the basic components of the digital circuit with one output and more than one input.

• AND, OR, and NOT gates are the basic gates.
• While NAND and NOR are the universal gates.
• EX-OR and EX-NOR are the special gates.

1. AND Gate:
The AND gate produces a 'high' output only when all its inputs are 'high'.



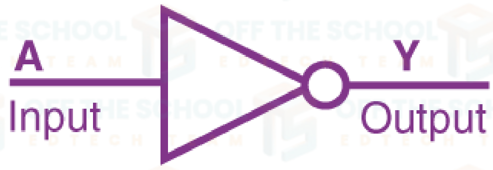
2. OR Gate:
An OR gate generates a 'high' output if any of its inputs are 'high'.



3. NOT Gate:

It is used to perform the inversion operation in digital circuits.

For example, the output of a NOT gate attains state 1 if and only if the input does not attain state 1.



4. NAND Gate:

Similarly, a NAND gate is an AND gate followed by a NOT gate, yielding the opposite output of an AND gate.

5. NOR Gate:

A NOR gate is an OR gate followed by a NOT gate, providing the opposite result of an OR gate.

6. XOR Gate (Exclusive-OR Gate):

The XOR gate outputs a 'high' signal if the number of "high" inputs is odd.

In other words, the Input Output output of a two-input XOR gate attains state 1 if one B adds only input and attains state 1.

7. XNOR Gate (Exclusive-NOR Gate):

In the XNOR gate, the output is in state 1 when both inputs are the same, that is, both 0 or both 1.